

HORMONAL FACTORS RELATED TO ABNORMAL GROWTH OF THE PROSTATE*
CHAPTER V

In view of the postulated hormone dependency of benign prostatic hyperplasia (BPH) and prostatic carcinoma (PCA), it is of importance to have an exact knowledge of the hormonal environment in which these tumors develop and to investigate whether in patients with either BPH or prostatic cancer, plasma sex hormone levels are different from those in controls.

This raises however some difficult problems. First, we really do not know when BPH or carcinoma really begins. Many urologists believe that a small periurethral nodule requires 15-25 years before it is transformed into clinical BPH (2,10). If BPH does not occur when castration is performed before puberty, but does occur if castration is performed between age 20 and 30, then it may be a valid working hypothesis that BPH begins early in adulthood. Second, if prepubertal castrates do not develop BPH even when treated with androgens, then it may be assumed that besides testosterone (T), the testes contribute (an) other factor(s), important in the development of BPH.

Hence, it appears that in evaluating the hormonal environment in which BPH develops, we should:

1. study sex hormone secretion not only in elderly males but also in young individuals.
2. not only look at testosterone but also at other testicular hormones.
3. not only make cross-sectional studies, but also make longitudinal studies over 30-40 years.

It is unrealistic however to hope that such data will be available in the next few years; hence, we are bound for many more years to rely upon cross-sectional studies. Finally, in view of the high incidence of BPH and PCA in elderly males, the rationale for comparing hormonal levels in patients with clinical prostatic neoplasia to levels in elderly males may be questioned.

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I. Sex hormone levels in normal males

1. After puberty plasma T levels reach a plateau with individual values between 280 and 1000 ng/dl, and a mean of about 600 ng/dl. The mean metabolic clearance rate (MCR) is about 1000 liters/24 hrs and the blood production rate (BPR) varies between 4 and 10 mg/24 hrs.

This testosterone is largely bound to plasma proteins: about 50-60% is bound to a high affinity, low capacity specific binding globulin, variously called testosterone-estradiol binding globulin (TeBG) or sex hormone binding globulin (SHBG); approximately 40% is loosely bound to albumin and 2% is free. Only the non specifically bound testosterone is biologically active. The albumin bound fraction is linearly related to the free fraction, the proportionality factor being determined only by the albumin concentration in plasma.

From the sixth decade of life on, there occurs a gradual decrease in plasma T levels, with however broad interindividual variations, and in the ninth decade of life the mean plasma T level is $\pm$ 250 ng/dl with individual levels as high as 600 or as low as 50 ng/dl.

Due to the progressive increase with age of the TeBG concentration, the free testosterone concentration starts to decrease already in the 5th decade of life and the combined effect of a decrease of the total T levels and of the increase in TeBG results in a steeper decline with age of free testosterone.

2. Total and/or free dihydrotestosterone (DHT) levels also decrease with age (21), but the decrease is by far less important than for testosterone (31,6,15).
3. Similarly, 5 α -androstane-3 α ,17 β -diol levels decrease from a mean of 17ng/dl in normal young males to a value of approximately 8ng/dl in elderly males.
4. Plasma androstenedione (A) has a mixed adrenal and testicular origin (31). Although Pirke et al (21) observed a moderate decrease with age of A levels, the relative levels of A are independent of age.
5. 17-hydroxy Progesterone (17-OHP) has essentially a testicular origin. Hence it is not surprising that an age dependent decrease of 17-OHP levels is observed.
6. Progesterone levels are age independent.

7. Along the $\Delta 5$ pathway, the immediate precursor of testosterone is androst-5-en-3 β ,17 β -diol ($\Delta 5$ -diol) (see chapter 2). Pirke et al (21) as well as we ourselves observed an age dependent decrease of plasma levels of $\Delta 5$ -diol.
8. The immediate biosynthetic precursor of $\Delta 5$ -diol is DHEA, plasma levels of which originate largely from the adrenal cortex. Considering its adrenal origin, plasma levels of DHEA as well as of DHEA-sulfate surprisingly decrease rapidly in elderly males (and women). In view of the fact that cortisol secretion does not decrease significantly in old age, this observation suggests a specific shift in the biosynthetic pathways in aging adrenals. Indeed, the decrease in plasma levels is not a consequence of a change in metabolism, as the MCR is certainly not increased in elderly persons.
9. Plasma 17-hydroxypregnenolone has a mixed adrenal-testicular origin with about equal contribution from both glands.

Again in elderly males the plasma levels are significantly lower than in young males; the same applies to pregnenolone levels.

From these results we may conclude that in aging males secretion of T and its precursors is decreased and that moreover there occur important changes in adrenal steroid biosynthesis, resulting in a dramatic decrease in DHEA and DHEA-sulfate levels.

What is the biological origin of this decrease in androgen secretion? Several morphological studies have shown that the biochemical evidence for an age dependent impairment of testicular function finds its anatomical equivalent in a decrease in testicular volume, progressive thickening of basal membranes, interstitial fibrosis, decreased spermatogenesis and a decrease in the number of Leydig cells. It has been shown that the distribution pattern of the testicular lesions is closely related to testicular blood flow, being more pronounced in the testicular segments at the periphery of arterial supply.

This raises the question whether testicular senescence has a primary testicular or a secondary hypothalamo-pituitary origin.

It is well known that, whereas in postmenopausal females there occurs a large increase in gonadotropin secretion, not nearly as great an increase is observed in elderly men.

Nevertheless, most authors studying plasma gonadotropin levels observed a

significant increase in both LH and FSH levels in old age, pointing towards a primary testicular origin of the decreased testicular function (27,23,8,13, 26).

Nevertheless there appears to exist an adequate Leydig cell secretory reserve since upon human chorionic gonadotropin stimulation plasma T levels increase, although the absolute and possibly also the relative increase is less than in younger subjects (23). On the other hand the pituitary gonadotrophs react to gonadotropin releasing hormone (GnRH) stimulation with increased LH and FSH output, although again the response may be somewhat weaker than in young individuals (23,8,9). Therefore, subtle age related alterations in the hypothalamo-pituitary function may exist with the resultant lowering of the feedback set point.

However, other explanations may be advanced. It is well known that in males estradiol (E2) levels do not decrease as a function of age and several authors observed an increase in E2 levels in elderly males (21,23). Moreover, due to the increase in TeBG and the lower affinity of TeBG for E2 than for T, the free E2 fraction decreases less than the free T fractions in elderly males. As a consequence there occurs an age related increase in the ratio of the concentration of available free estradiol to available free testosterone (AFE2C/AFTC). Hence it is possible that this increased E2 level might suppress LH secretion which normally would be stimulated in response to the age related decrease in T levels.

II. Sex hormone levels in BPH and prostatic carcinoma

The question may be raised whether subjects with BPH or prostatic carcinoma have a different hormonal spectrum. Schematically aging is characterized by a decline in androgen secretion and at least a relative increase in estrogen secretion.

In view of the fact that androgens and perhaps estrogens are necessary for BPH to develop, one may wonder in subjects that develop BPH, whether higher androgen and/or estrogen levels are found. The difficulty however resides in the fact:

1. that BPH becomes only clinically manifest after a silent evolution of at least 20 yrs.
2. that clinical symptomatology is only poorly related to the volume of the prostate.

3. that at the time of clinical symptomatology, the critical period for growth stimulation may be over.

As no prospective studies are available and as subjects that will develop BPH or prostatic carcinoma cannot be distinguished early in life, the only data available concern patients in an advanced stage of the disease. There is some evidence for distinct differences in hormonal plasma levels between patients with clinically evident BPH and "normal controls".

1. In comparison to normal controls, BPH patients appear to have higher mean plasma T (fig. 1) and DHT levels (11,12,29,30); the differences being greatest within patients 70-80 yrs old. The individual values overlap, however, the whole normal range. In PCA the mean T level, although higher, was not significantly increased in comparison to control.
2. Whereas in young males, 75% of plasma DHT originates by peripheral conversion of T, in elderly males (with BPH?) over 50% of DHT is secreted as such either by the testes or the prostate.

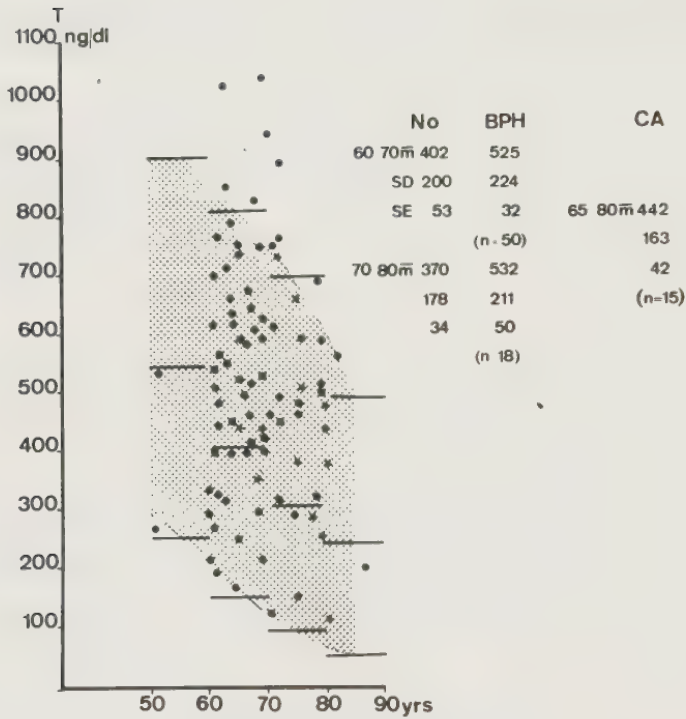
It is possible that the hypertrophied prostate secretes DHT. Mahoudeau et al. (17) found the mean DHT levels in prostatic venous plasma to be 69 vs 41 ng/dl in peripheral plasma, although the difference was not statistically significant. However, if we estimate the prostatic blood flow at about 20 ml/min, the secretion of the hyperplastic prostate would be $\pm 10 \mu\text{g}/24 \text{ hrs}$ or relatively negligible.

After prostatectomy neither Mahoudeau et al. (17) nor Vermeulen & De Sy (30) found any decrease in DHT levels, suggesting again that prostatic DHT secretion does not contribute significantly to the total DHT blood production rate ($300-600 \mu\text{g}/24 \text{ hrs}$).

3. Mean plasma estradiol levels, ($3.0 \pm 0.2 \text{ ng/dl}$) appear to be slightly higher in BPH patients than in controls ($2.2 \pm 0.2 \text{ ng/dl}$), however with individual values overlapping again the whole normal range.(Fig.2) This confirms the data of Sköldfors et al. (24) which demonstrated an increase in urinary excretion of estrogens in BPH. In this connection it is interesting to recall that Lloyd et al. (16) reported that in dogs with BPH, E2 and E1 plasma and prostatic tissue levels were significantly increased.

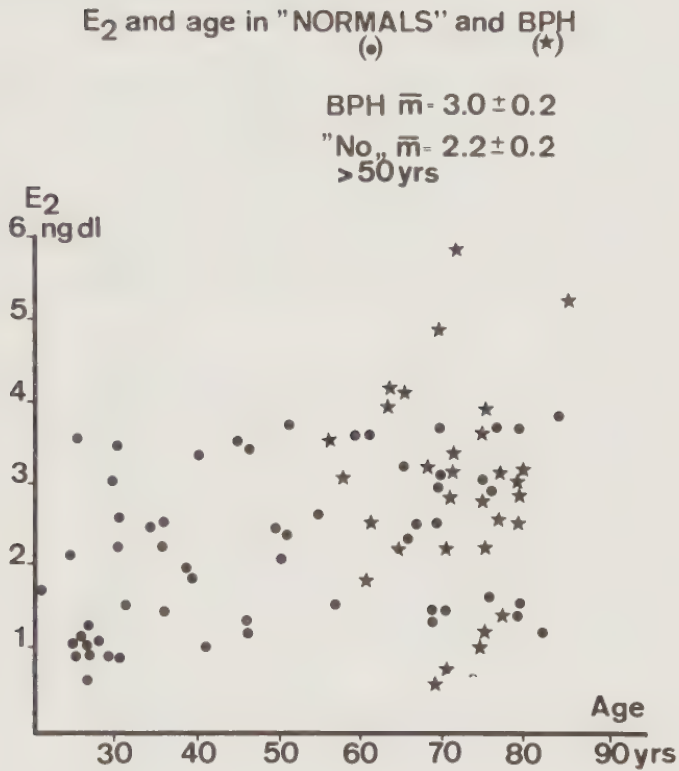
Bayard et al. (1) on the other hand found similar T and E2 blood

Figure 1.



T levels in function of age, in normal subjects (No) and in patients with prostatic neoplasm. • = BPH * = PCA

Figure 2.



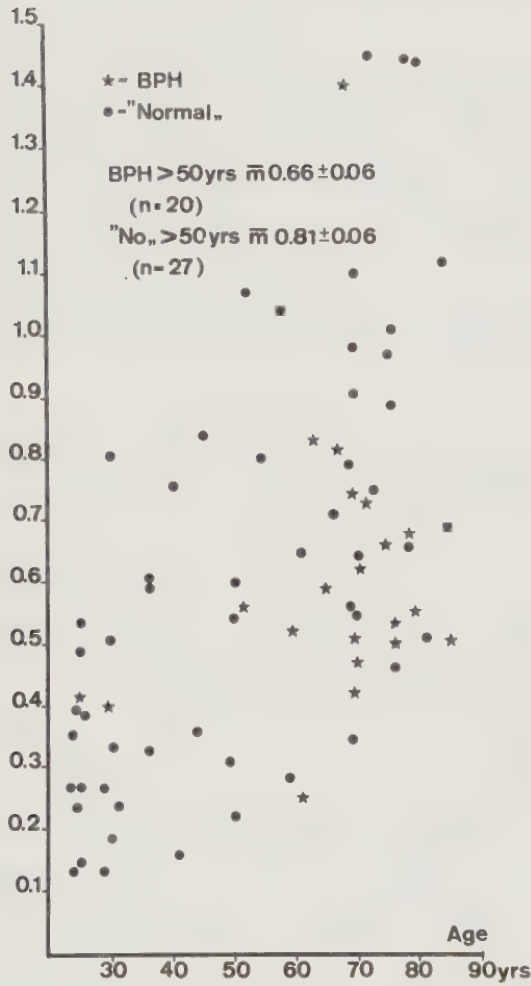
Estradiol levels in aging males with and without BPH.

production rates in 6 patients with BPH and in sex age matched controls.

4. As shown in fig. 3 there is an evident increase of the E2/T ratio with age but in patients with clinical BPH, this ratio is not higher than in "normal" controls.
5. In patients with either BPH or prostatic carcinoma, LH but not FSH levels appear to be lower than in controls (7), as is the increase upon GnRH stimulation (4,7).
6. As far as prolactin is concerned, data in the literature are divergent as Harper et al. observed increased levels in patients with BPH, whereas Hammond et al. (7) did not observe any difference with normal controls.

In summary, data from literature as well as from our laboratory suggest that in patients with clinical BPH and possibly also with prostatic carcinoma, both mean androgen and estrogen levels, are slightly higher than in normal controls. Moreover there is evidence for at least relatively increased secretion of DHT, but available data so far do not indicate that the prostate is a major source of the extra DHT. In the individual patient with BPH however, low, normal or increased androgen and/or estrogen levels may be observed. As to the significance of these findings with respect to pathogenesis of either BPH or prostatic carcinoma, one can only speculate. Androgens do stimulate prostatic growth and there is suggestive evidence that the combined action of androgens and estrogens might be synergistic. On the other hand increased plasma levels may be the consequence rather than the cause of prostatic neoplasia. For example, in BPH tissue the conversion of DHT to Adiol could be impaired (5,20); this might block further uptake of DHT, and although rather unlikely, increase indirectly plasma DHT levels. Malathi & Gurpide (18) in superfusion experiments found that at least under in vitro conditions release of DHT to cell medium is a main route of disposition of DHT, whereas most of Adiol and T are metabolized to other products. Wilson et al. (33) were unable to produce BPH with injection of either T or DHT in castrated dogs, although the intracellular DHT concentration was as high as in the control hypertrophic group. Androstane-3 α ,17 β -diol, alone or in combination with estrogens on the other hand did induce BPH in castrated dogs (32) and, in distinction with other authors Shida et al. (25) observed an increased Adiol formation in hypertrophic in comparison to normal prostates (14). On the

Figure 3.



E2/T in males with and without BPH.

other hand, in the view of the low Adiol concentration in plasma, its decrease in old age and its high affinity binding to TeBG, it appears however difficult to conceive how plasma Adiol would play a major role in the pathogenesis of BPH.

One should finally bear in mind that all the data concerning hormone levels in prostatic neoplasia are obtained at the time when BPH or PCA are well developed, and that their developments took 10-20 yrs. We do not know whether patients in whom increased hormone levels are observed have had increased hormone levels for the whole period while they developed prostatic neoplasia. Prospective longitudinal studies in young subjects with elevated sex hormone levels should furnish data to decide whether increased sex hormone levels play a causal role in the pathogenesis of prostatic neoplasia.

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